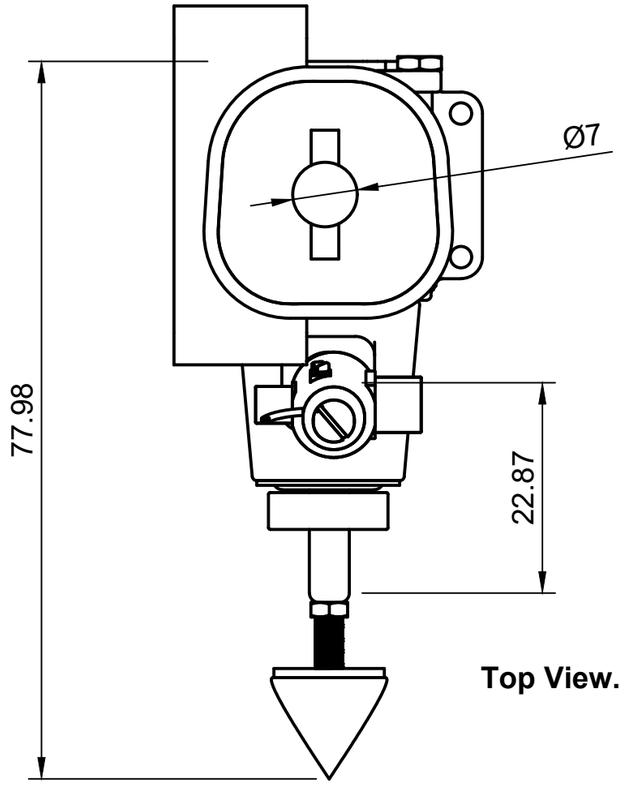
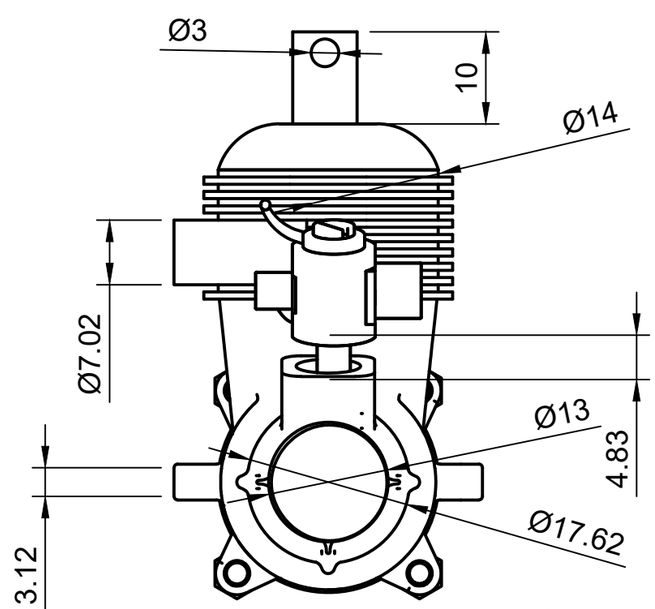
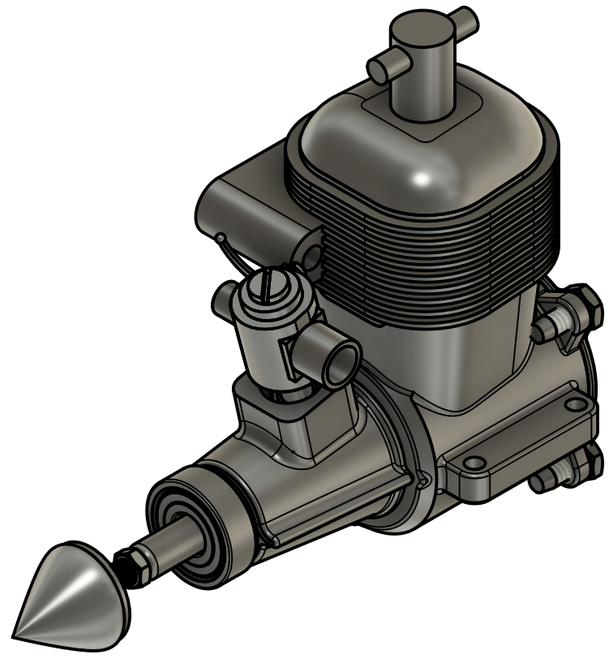
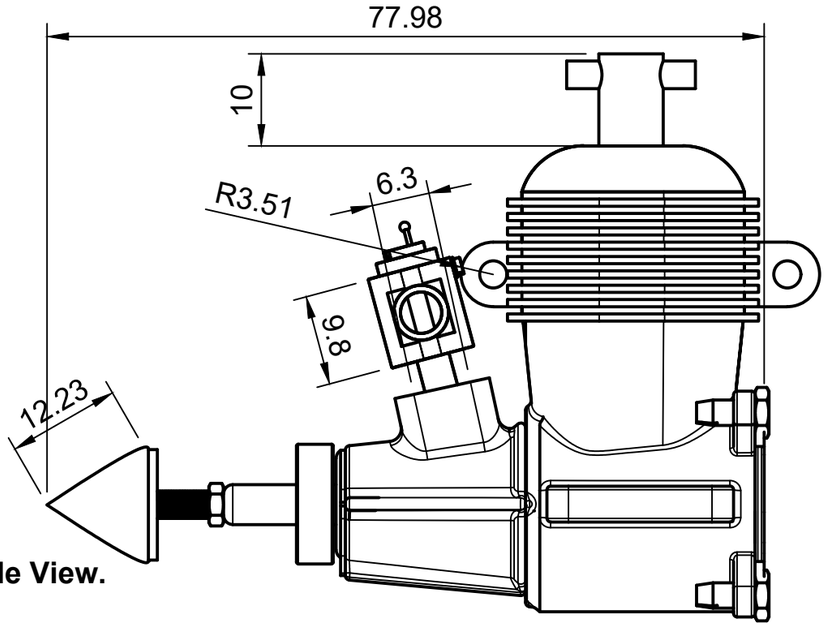


Top View.



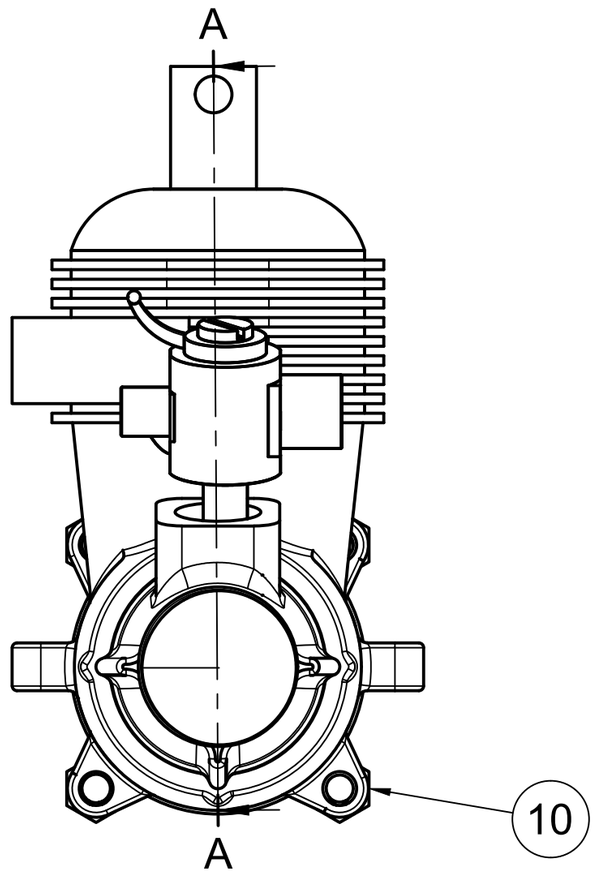
Front View.



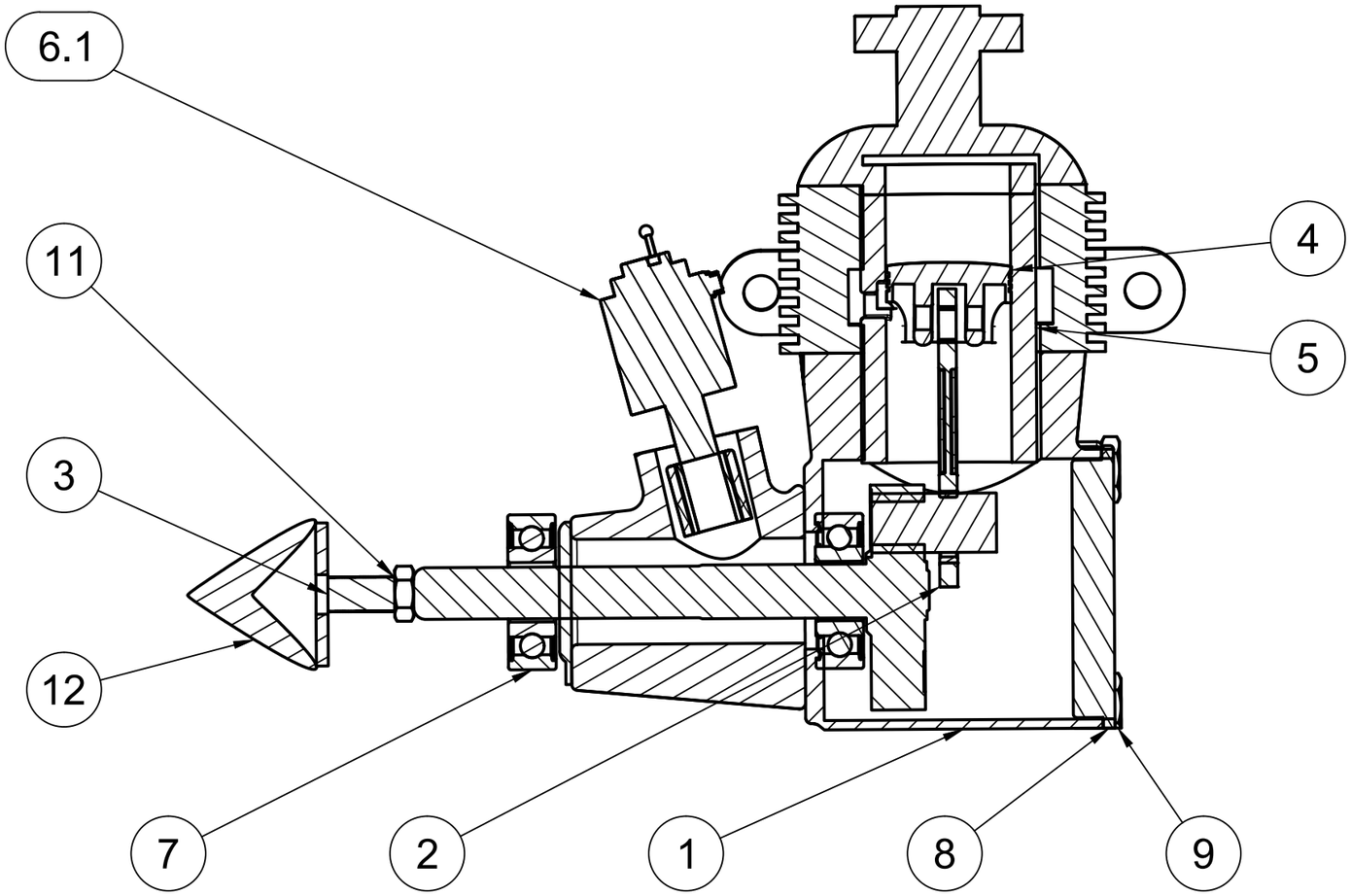


Side View.

Third Angle Projection of Engine Assembly.	
Group No.: No. 9	University of Birmingham.
Scale 2:1	*Material:* AISI 4340 Steel.
Mechanical Design A.	*Date of Issue:* 16/12/25
Drawing to BS 8888.	*Drawing No.* 1/1



A-A (2:1)



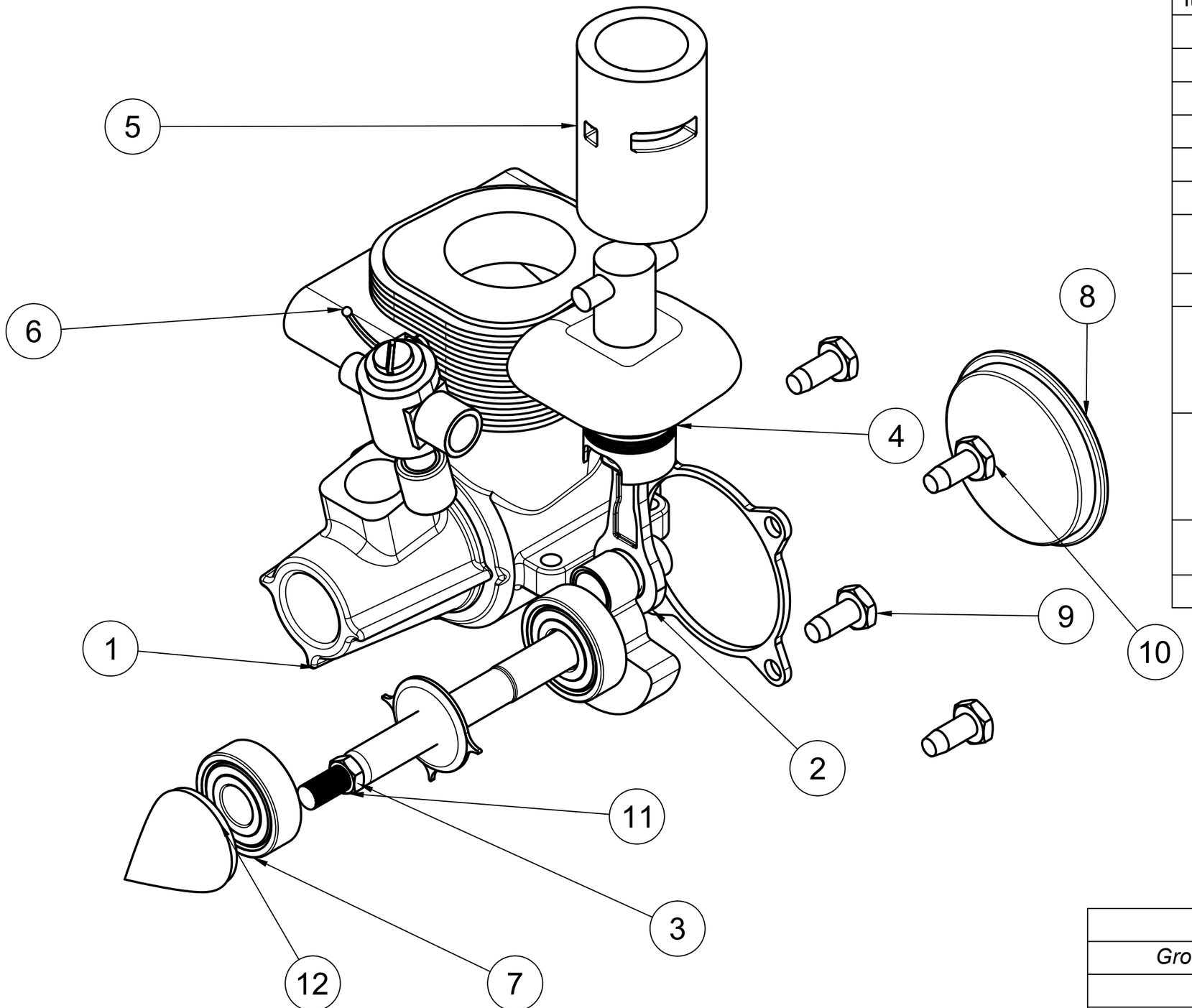
Item	Qty	Part Number	Material
8	1	Backplate	Steel
7	2	SKF_bearing_619_5-2Z_2_1	Steel
6.1	1	Component1	Steel
6	1	Carburetor new_	Steel
5	1	Cylinder liner	Steel
4	1	Piston Tarah	Steel
3	1	Contrapeso	Steel
2	1	Connecting rod	----
1	1	Casing.	Steel

Parts List

Item	Qty	Part Number	Material
12	1	Nose Cone Tarah	Steel
11	1	UNI EN 24032 - M2 Steel 6 Plain	Steel 6, Plain
10	2	PRHHTSTF ASME B18.6.4 - 4-40 UNC x 0.25 Steel Grade 2 Plain	Steel Grade 2, Plain
9	2	PRHHTSTF ASME B18.6.4 - 4-48 UNF x 0.25 Steel Grade 2 Plain	Steel Grade 2, Plain

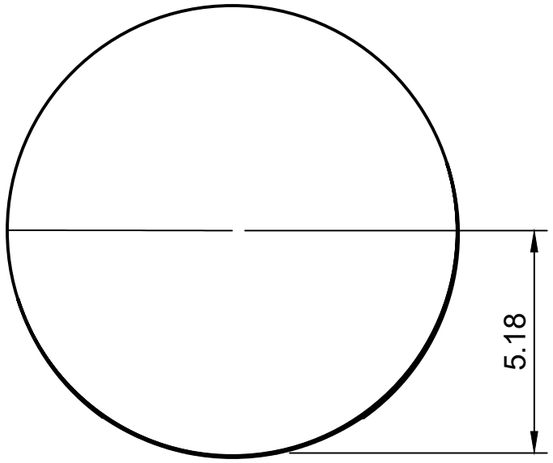
Parts List

Section View of Final Assembly.	
Group No.: No. 9	University of Birmingham.
Scale 2:1	*Material:* AISI 4340 Steel.
Mechanical Design A.	*Date of Issue:* 16/12/25
Drawing to BS 8888.	*Drawing No.* 1/1

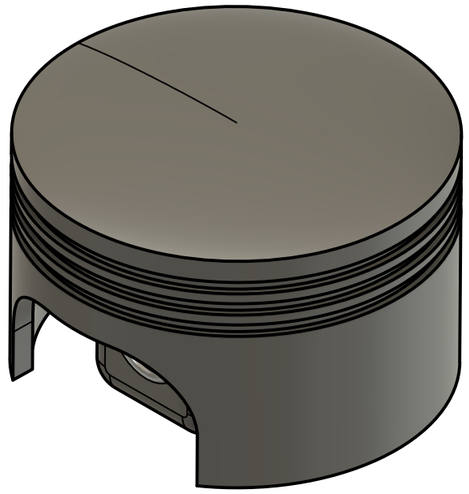


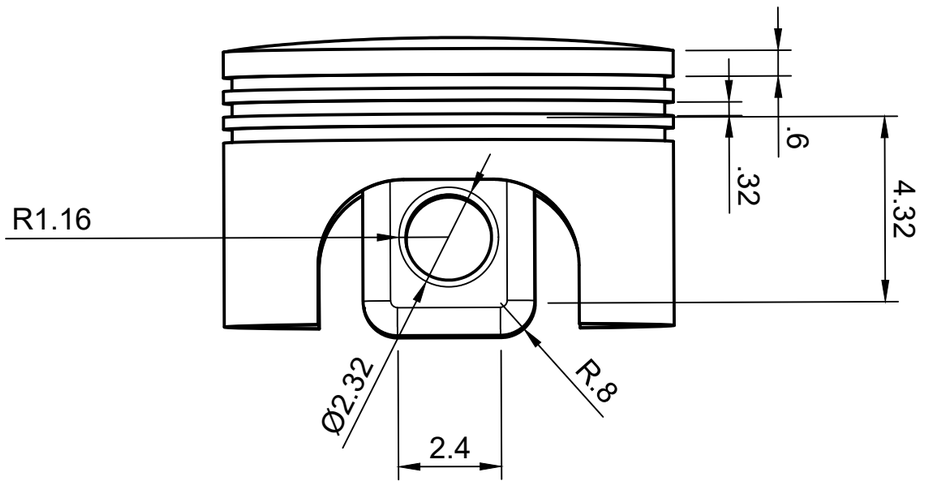
Parts List			
Item	Qty	Part Number	Material
1	1	Casing.	Steel
2	1	Connecting rod	----
3	1	Crankshaft	Steel
4	1	Piston head	Steel
5	1	Cylinder liner	Steel
6	1	Carburetor	Steel
7	2	SKF_bearing_619_5-2Z_2_1	Steel
8	1	Backplate	Steel
9	2	PRHHTSTF ASME B18.6.4 - 4-48 UNF x 0.25 Steel Grade 2 Plain	Steel Grade 2, Plain
10	2	PRHHTSTF ASME B18.6.4 - 4-40 UNC x 0.25 Steel Grade 2 Plain	Steel Grade 2, Plain
11	1	UNI EN 24032 - M2 Steel 6 Plain	Steel 6, Plain
12	1	Nose Cone	Steel

Exploded View of Final Assembly.	
Group No.: No. 9	University of Birmingham.
Scale 2:1	*Material:* AISI 4340 Steel.
Mechanical Design A.	*Date of Issue:* 16/12/25
Drawing to BS 8888.	*Drawing No.* 1/1

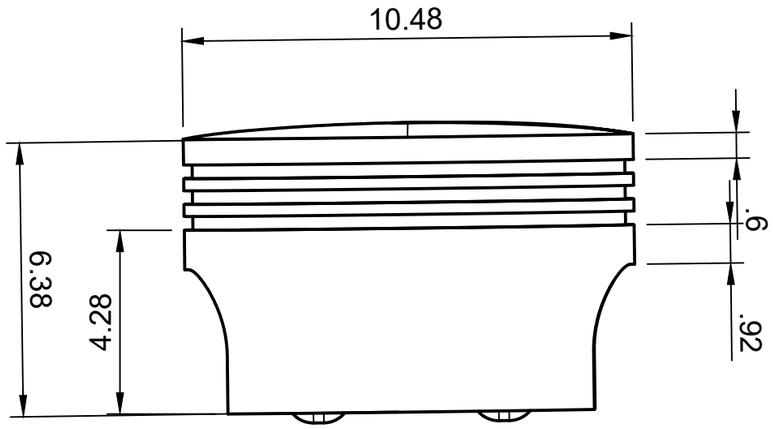


Top View.



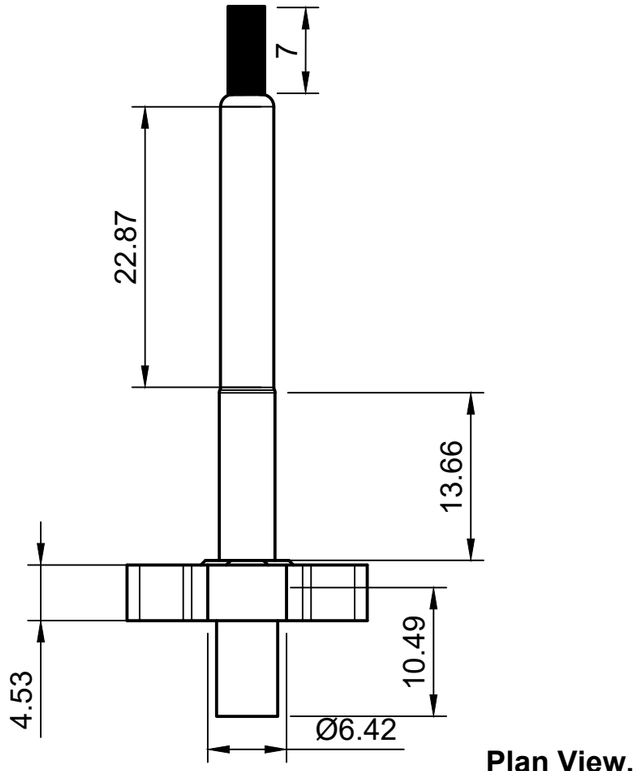


Front View.

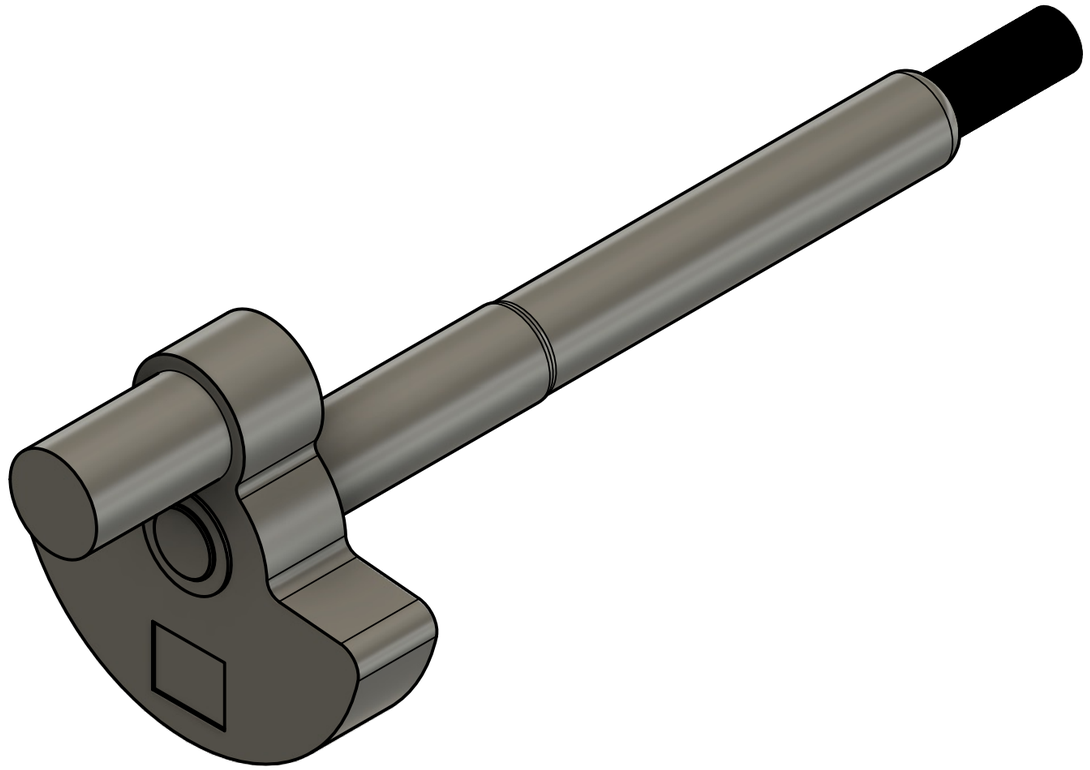


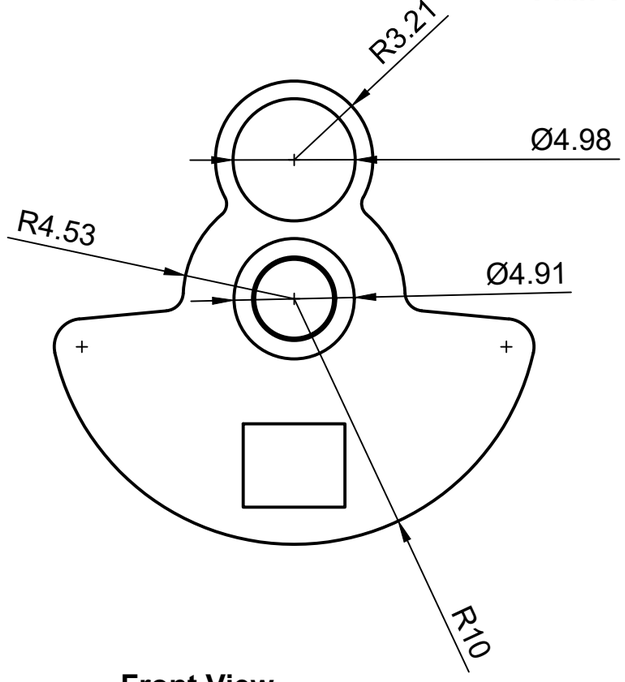
Side View.

Piston Head.	
Group No.: No. 9	University of Birmingham.
Scale 7:1	*Material:* AISI 4340 Steel.
Mechanical Design A.	*Date of Issue:* 16/12/25
Drawing to BS 8888.	*Drawing No.* 1/1

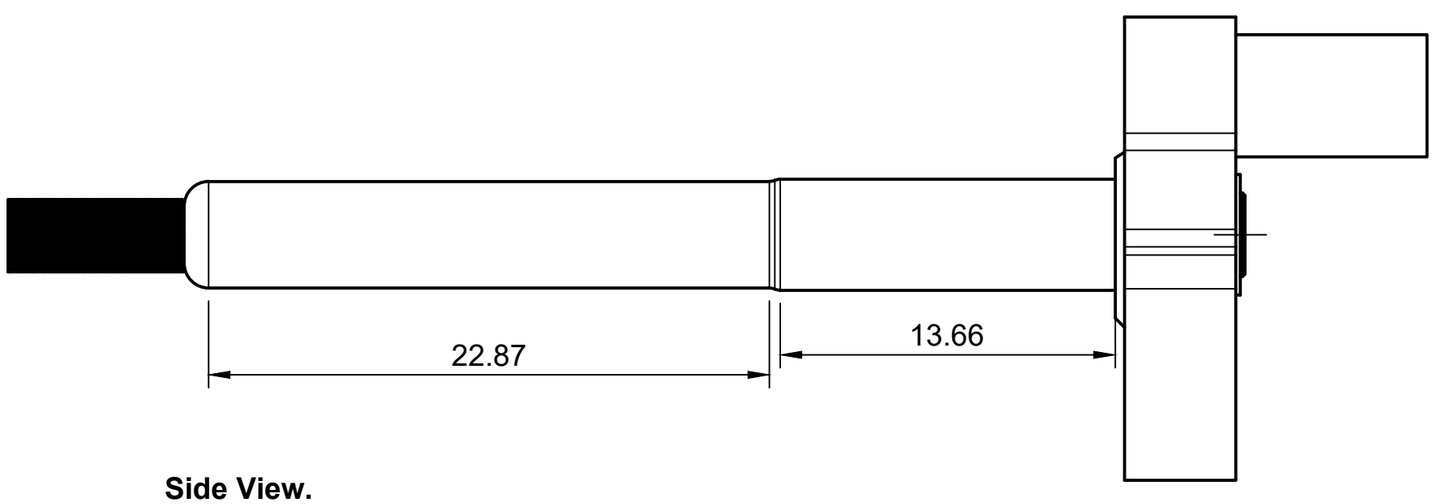


Plan View.



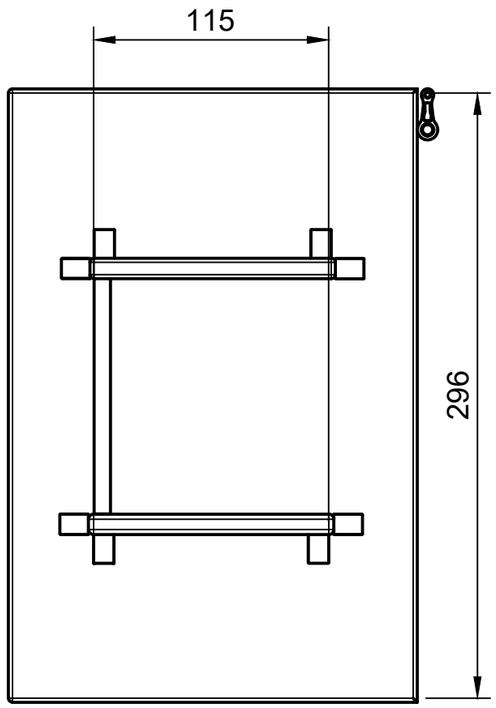


Front View.

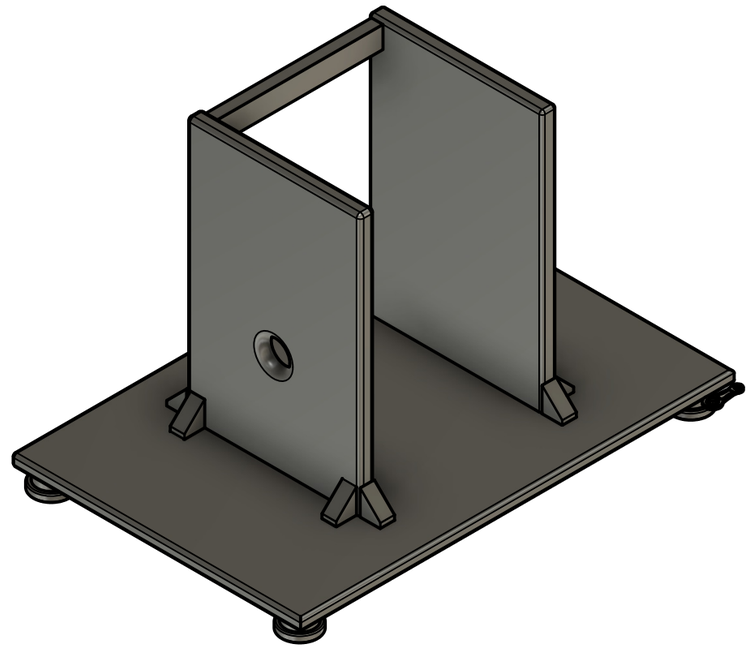


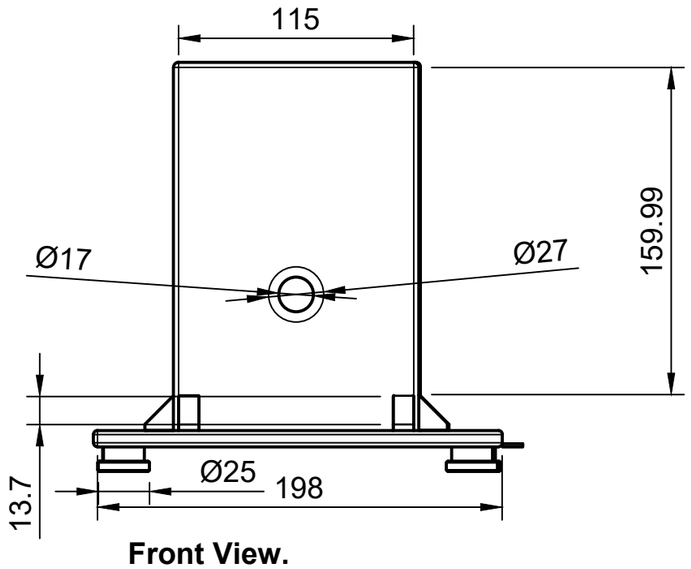
Side View.

Crankshaft drawing.	
Group No.: No. 9	University of Birmingham.
Scale 4:1	*Material:* AISI 4340 Steel.
Mechanical Design A.	*Date of Issue:* 16/12/25
Drawing to BS 8888.	*Drawing No.* 1/1

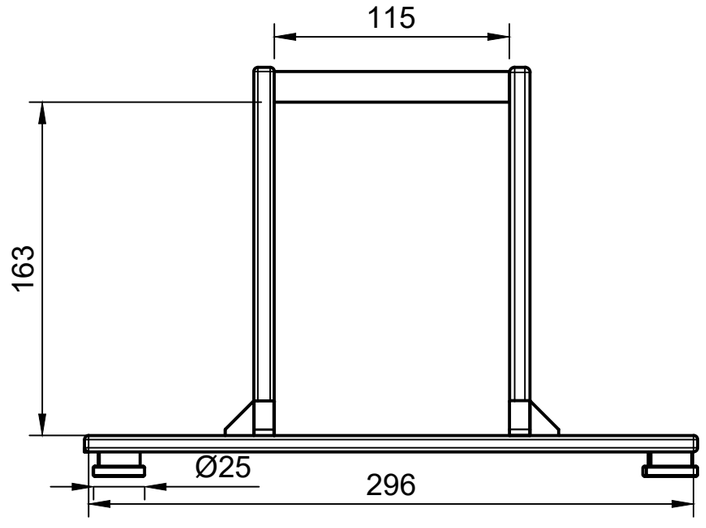


Top View.





Front View.



Side View.

Engine Holder Drawing.	
Group No.: No. 9	University of Birmingham.
Scale 1:3	*Material:* AISI 4340 Steel.
Mechanical Design A.	*Date of Issue:* 16/12/25
Drawing to BS 8888.	*Drawing No.* 1/1